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Last login: Sun Jun 28 06:12:46 on ttys000
stephenrajva@Stephens-MacBook-Air-4 ~ % cd "/Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/Mi
niFE/Mesh"
echo "==== folder contents ====="
ls -la
echo ""
echo "==== PPN=1 config (mesh_ppn1.py) ====="
cat mesh_ppn1.py 2>/dev/null || echo "mesh_ppn1.py NOT here - listing any .py files:"
ls *.py 2>/dev/null
==== folder contents =====
total 1120
drwxr-xr-x  15 stephenrajva  staff    480 24 Jun 18:40 .
drwxr-xr-x   9 stephenrajva  staff    288 24 Jun 23:05 ..
-rw-r--r--@  1 stephenrajva  staff  14340 28 Jun 06:08 .DS_Store
-rw-r--r--@  1 stephenrajva  staff   3943 24 Jun 08:00 analyze_mesh.py
-rw-r--r--@  1 stephenrajva  staff  54400 24 Jun 08:00 hop_count_vs_size.png
drwxr-xr-x   7 stephenrajva  staff    224 24 Jun 07:23 individual_files
-rw-r--r--@  1 stephenrajva  staff  62210 24 Jun 08:00 latency_vs_size.png
-rw-r--r--@  1 stephenrajva  staff   6706 24 Jun 01:40 mesh_ppn1.py
drwxr-xr-x   2 stephenrajva  staff     64 23 Jun 10:41 PPN_Study
drwxr-xr-x  11 stephenrajva  staff    352 25 Jun 10:10 ppn1
drwxr-xr-x  25 stephenrajva  staff    800 23 Jun 15:17 PPN1_Scaling
-rw-r--r--@  1 stephenrajva  staff   3938 24 Jun 01:40 README_PPN_study.md
drwxr-xr-x   7 stephenrajva  staff    224 24 Jun 08:16 results
-rw-r--r--@  1 stephenrajva  staff   1690 24 Jun 01:40 run_mesh_sweep.sh
-rw-r--r--@  1 stephenrajva  staff  411918 24 Jun 18:40 Screenshot 2026-06-24 at 6.40.29 PM.png

==== PPN=1 config (mesh_ppn1.py) =====
#!/usr/bin/env python
# =====
# mesh_ppn1.py - Mesh topology scaling study with PPN (ranks per node) = 1
# =====
# PPN = "Processes Per Node" = number of MPI ranks placed on each compute node.
#
# IMPORTANT: In SST there is NO parameter literally called "ppn".
# PPN is a DERIVED number:
#
#         PPN = (number of MPI ranks) / (number of compute nodes)
#               =      job["size"]      / (num_routers * local_ports)
#
# To FIX PPN = 1 we simply make:  job size == number of nodes.
# This script computes that automatically, so PPN is guaranteed to be 1.
#
# USAGE (one shape per run, chosen with an environment variable):
#     export MESH_SHAPE="4x4"          # router grid -> here 16 nodes
#     sst mesh_ppn1.py
#
# The runner script run_mesh_sweep.sh loops over all the shapes for you.
# =====

import sst
import os
from sst.merlin.base import *
from sst.merlin.endpoint import *
from sst.merlin.interface import *
from sst.merlin.topology import *
from sst.ember import *

# -----
# 1. STUDY PARAMETERS (the only things that change between runs)
# -----
PPN = 1 # <-- FIXED at 1 (ranks per node)
shape = os.environ.get("MESH_SHAPE", "4x4") # router grid, e.g. "2x2", "8x8"
nodes_per_router = 1 # local_ports = 1 node per router
out_dir = os.environ.get("STATS_DIR", ".") # where to drop the CSV

# Work out how many routers / nodes / ranks this shape implies
dims = [int(x) for x in shape.lower().split("x")]
num_routers = 1
for d in dims:
    num_routers *= d
num_nodes = num_routers * nodes_per_router # total compute nodes (endpoints)
num_ranks = num_nodes * PPN # PPN=1 -> ranks == nodes

# -----
# 2. PLATFORM + TOPOLOGY

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# -----
PlatformDefinition.setCurrentPlatform("firefly-defaults")

topo                = topoMesh()
topo.shape          = shape
topo.local_ports    = nodes_per_router          # nodes hanging off each router
topo.width          = "x".join(["1"] * len(dims)) # 1 link per dimension = clean mesh
topo.link_latency   = "25ns"

# -----
# 3. ROUTER
# -----
router              = hr_router()
router.link_bw      = "12GB/s"
router.flit_size    = "16B"
router.xbar_bw      = "20GB/s"
router.input_latency = "30ns"
router.output_latency = "30ns"
router.input_buf_size = "32kB"
router.output_buf_size = "32kB"
router.num_vns      = 2
router.xbar_arb      = "merlin.xbar_arb_lru"
topo.router = router

# -----
# 4. NETWORK INTERFACE (one NIC per node)
# -----
nic                = ReorderLinkControl()
nic.link_bw        = "12GB/s"
nic.input_buf_size = "16kB"
nic.output_buf_size = "16kB"

# -----
# 5. SYSTEM
# -----
system = System()
system.setTopology(topo)

# -----
# 6. WORKLOAD (MiniFE-style collectives + a few representative point-to-point)
# -----
motifs = [
    {"pattern": "Allreduce", "params": "iterations=92 count=1 compute=69ns"},
    {"pattern": "Bcast",      "params": "root=0 iterations=2 count=7 compute=1323ns"},
    {"pattern": "Reduce",     "params": "root=0 iterations=2 count=2 compute=450ns"},
    {"pattern": "Allgather",  "params": "iterations=3 count=3 compute=243ns verify=0"},
    {"pattern": "Barrier",    "params": "iterations=99 compute=75ns"},
]

# Pick PingPong partners that ALWAYS exist for this size (rank2 must be < num_ranks).
# We sample a near neighbour, a one-row-down neighbour, the middle, and the far corner.
row_step = dims[-1] if len(dims) >= 2 else 1
candidates = [1, row_step, num_nodes // 2, num_nodes - 1]
partners = sorted({p for p in candidates if 0 < p < num_ranks})

for b in partners:
    motifs.append(
        {"pattern": "PingPong",
         "params": f"rank2={b} iterations=35 messageSize=256 compute=3ns"}
    )

# -----
# 7. BUILD THE JOB (size == num_nodes => PPN guaranteed = 1)
# -----
ep = EmberMPIJob(0, num_ranks)
ep.network_interface = nic
ep.addMotif("Init")
for m in motifs:
    ep.addMotif(f"{m['pattern']} {m['params']}")
ep.addMotif("Fini")
ep.nic.nic2host_lat = "100ns"

system.allocateNodes(ep, "linear")
system.build()

# -----

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# 8. STATISTICS -> one CSV per shape
# -----
csv_path = os.path.join(out_dir, f"mesh_{shape}_ppn{PPN}_nodes{num_nodes}_stats.csv")

sst.setStatisticLoadLevel(10)
sst.setStatisticOutput("sst.statOutputCSV", {"filepath": csv_path, "separator": ", "})

sst.enableAllStatisticsForComponentType("merlin.hr_router")
sst.enableAllStatisticsForComponentType("merlin.linkcontrol")
sst.enableAllStatisticsForComponentType("ember.EmberEngine", {"MotifLog": "1"})
sst.enableAllStatisticsForComponentType("ember.nic")

# -----
# 9. CONFIG SUMMARY (prints to terminal so you can verify PPN at a glance)
# -----
print("=====")
print(f" Mesh shape      : {shape}")
print(f" Routers         : {num_routers}")
print(f" Nodes per router : {nodes_per_router} (local_ports)")
print(f" Total nodes      : {num_nodes}")
print(f" Total MPI ranks  : {num_ranks}")
print(f" PPN (ranks/node) : {num_ranks // num_nodes} <-- should be 1")
print(f" Stats CSV        : {csv_path}")
print("=====")
analyze_mesh.py mesh_ppn1.py
stephenrajva@Stephens-MacBook-Air-4 Mesh % cd "/Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/MiniFE/Mesh"
mkdir -p PPN_2
find ~/Downloads -name "mesh_ppn2.py" 2>/dev/null
find ~/Downloads -name "run_mesh_sweep_ppn2.sh" 2>/dev/null
/Users/stephenrajva/Downloads/mesh_ppn2.py
/Users/stephenrajva/Downloads/run_mesh_sweep_ppn2.sh
stephenrajva@Stephens-MacBook-Air-4 Mesh % cd "/Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/MiniFE/Mesh/PPN_2"
export MESH_SHAPE="2x2"
export STATS_DIR="./results"
sst mesh_ppn2.py
=====
 Mesh shape      : 2x2
 Routers         : 4
 Nodes per router : 2 (local_ports)
 Total nodes      : 8
 Total MPI ranks  : 8
 PPN (ranks/node) : 2 <-- should be 2
 Stats CSV        : ./results/mesh_2x2_ppn2_nodes8_stats.csv
=====
Allreduce: ranks 8, loop 92, 1 double(s), latency 5.507 us
generate
510120733:0:0:EmberEngine:Reduce:generate:37: rank=0 size=8
generate
511463933:0:1:EmberEngine:Reduce:generate:37: rank=1 size=8
generate
511679933:0:2:EmberEngine:Reduce:generate:37: rank=2 size=8
generate
512263133:0:3:EmberEngine:Reduce:generate:37: rank=3 size=8
generate
512263533:0:4:EmberEngine:Reduce:generate:37: rank=4 size=8
generate
512435533:0:5:EmberEngine:Reduce:generate:37: rank=5 size=8
generate
512479533:0:6:EmberEngine:Reduce:generate:37: rank=6 size=8
generate
Bcast: ranks 8, loop 2, bytes 56, latency 1.893 us
513062733:0:7:EmberEngine:Reduce:generate:37: rank=7 size=8
Allgather: ranks 8, loop 3, bytes 12, latency 5.041 us
Barrier: ranks 8, loop 99, latency 5.506 us
1099539533:0:0:EmberEngine:PingPong:generate:65: rank=0 size=8
rank2=1 messageSize=256 iterations=35
1100382733:0:1:EmberEngine:PingPong:generate:65: rank=1 size=8
1100598733:0:2:EmberEngine:PingPong:generate:65: rank=2 size=8
1101181933:0:4:EmberEngine:PingPong:generate:65: rank=4 size=8
1101981933:0:7:EmberEngine:PingPong:generate:65: rank=7 size=8
PingPong: otherRank 1, total time 63.474 us, loop 35, bufLen 256, latency 0.907 us. bandwidth 0.282320 G
B/s
1163013133:0:0:EmberEngine:PingPong:generate:65: rank=0 size=8
rank2=2 messageSize=256 iterations=35

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PingPong: otherRank 2, total time 69.272 us, loop 35, bufLen 256, latency 0.990 us. bandwidth 0.258690 G
B/s
1232285133:0:0:EmberEngine:PingPong:generate:65: rank=0 size=8
rank2=4 messageSize=256 iterations=35
PingPong: otherRank 4, total time 69.272 us, loop 35, bufLen 256, latency 0.990 us. bandwidth 0.258690 G
B/s
1301557133:0:0:EmberEngine:PingPong:generate:65: rank=0 size=8
rank2=7 messageSize=256 iterations=35
PingPong: otherRank 7, total time 75.320 us, loop 35, bufLen 256, latency 1.076 us. bandwidth 0.237918 G
B/s
Simulation is complete, simulated time: 1.38146 ms
stephenrajva@Stephens-MacBook-Air-4 PPN_2 % cd "/Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/MiniFE/Mesh/PPN_2"
./run_mesh_sweep_ppn2.sh
zsh: permission denied: ./run_mesh_sweep_ppn2.sh
stephenrajva@Stephens-MacBook-Air-4 PPN_2 % cd "/Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/MiniFE/Mesh/PPN_2"
chmod +x run_mesh_sweep_ppn2.sh
./run_mesh_sweep_ppn2.sh
=== MiniFE mesh 2x2 (PPN=2) ===
=====
Mesh shape      : 2x2
Routers         : 4
Nodes per router : 2 (local_ports)
Total nodes     : 8
Total MPI ranks  : 8
PPN (ranks/node) : 2  <-- should be 2
Stats CSV       : /Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/MiniFE/Mesh/PPN_2/results
/mesh_2x2_ppn2_nodes8_stats.csv
=====
Allreduce: ranks 8, loop 92, 1 double(s), latency 5.507 us
generate
510120733:0:0:EmberEngine:Reduce:generate:37: rank=0 size=8
generate
511463933:0:1:EmberEngine:Reduce:generate:37: rank=1 size=8
generate
511679933:0:2:EmberEngine:Reduce:generate:37: rank=2 size=8
generate
512263133:0:3:EmberEngine:Reduce:generate:37: rank=3 size=8
generate
512263533:0:4:EmberEngine:Reduce:generate:37: rank=4 size=8
generate
512435533:0:5:EmberEngine:Reduce:generate:37: rank=5 size=8
generate
512479533:0:6:EmberEngine:Reduce:generate:37: rank=6 size=8
generate
Bcast: ranks 8, loop 2, bytes 56, latency 1.893 us
513062733:0:7:EmberEngine:Reduce:generate:37: rank=7 size=8
Allgather: ranks 8, loop 3, bytes 12, latency 5.041 us
Barrier: ranks 8, loop 99, latency 5.506 us
1099539533:0:0:EmberEngine:PingPong:generate:65: rank=0 size=8
rank2=1 messageSize=256 iterations=35
1100382733:0:1:EmberEngine:PingPong:generate:65: rank=1 size=8
1100598733:0:2:EmberEngine:PingPong:generate:65: rank=2 size=8
1101181933:0:4:EmberEngine:PingPong:generate:65: rank=4 size=8
1101981933:0:7:EmberEngine:PingPong:generate:65: rank=7 size=8
PingPong: otherRank 1, total time 63.474 us, loop 35, bufLen 256, latency 0.907 us. bandwidth 0.282320 G
B/s
1163013133:0:0:EmberEngine:PingPong:generate:65: rank=0 size=8
rank2=2 messageSize=256 iterations=35
PingPong: otherRank 2, total time 69.272 us, loop 35, bufLen 256, latency 0.990 us. bandwidth 0.258690 G
B/s
1232285133:0:0:EmberEngine:PingPong:generate:65: rank=0 size=8
rank2=4 messageSize=256 iterations=35
PingPong: otherRank 4, total time 69.272 us, loop 35, bufLen 256, latency 0.990 us. bandwidth 0.258690 G
B/s
1301557133:0:0:EmberEngine:PingPong:generate:65: rank=0 size=8
rank2=7 messageSize=256 iterations=35
PingPong: otherRank 7, total time 75.320 us, loop 35, bufLen 256, latency 1.076 us. bandwidth 0.237918 G
B/s
Simulation is complete, simulated time: 1.38146 ms
=== MiniFE mesh 4x4 (PPN=2) ===
=====
Mesh shape      : 4x4
Routers         : 16
Nodes per router : 2 (local_ports)

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Total nodes : 32
Total MPI ranks : 32
PPN (ranks/node) : 2 <-- should be 2
Stats CSV : /Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/MiniFE/Mesh/PPN_2/results
/mesh_4x4_ppn2_nodes32_stats.csv

=====
Allreduce: ranks 32, loop 92, 1 double(s), latency 9.587 us
generate
885489533:0:0:EmberEngine:Reduce:generate:37: rank=0 size=32
generate
886832733:0:1:EmberEngine:Reduce:generate:37: rank=1 size=32
generate
887048733:0:2:EmberEngine:Reduce:generate:37: rank=2 size=32
generate
887762333:0:3:EmberEngine:Reduce:generate:37: rank=3 size=32
generate
887978333:0:5:EmberEngine:Reduce:generate:37: rank=5 size=32
generate
887987733:0:4:EmberEngine:Reduce:generate:37: rank=4 size=32
generate
888194933:0:6:EmberEngine:Reduce:generate:37: rank=6 size=32
generate
888778333:0:7:EmberEngine:Reduce:generate:37: rank=7 size=32
generate
888908533:0:8:EmberEngine:Reduce:generate:37: rank=8 size=32
generate
888994333:0:11:EmberEngine:Reduce:generate:37: rank=11 size=32
generate
889038133:0:12:EmberEngine:Reduce:generate:37: rank=12 size=32
generate
889090133:0:9:EmberEngine:Reduce:generate:37: rank=9 size=32
generate
889133933:0:10:EmberEngine:Reduce:generate:37: rank=10 size=32
generate
889210933:0:13:EmberEngine:Reduce:generate:37: rank=13 size=32
generate
889254733:0:14:EmberEngine:Reduce:generate:37: rank=14 size=32
generate
889577533:0:15:EmberEngine:Reduce:generate:37: rank=15 size=32
generate
889577933:0:17:EmberEngine:Reduce:generate:37: rank=17 size=32
generate
889794733:0:18:EmberEngine:Reduce:generate:37: rank=18 size=32
generate
889836333:0:23:EmberEngine:Reduce:generate:37: rank=23 size=32
generate
889845933:0:19:EmberEngine:Reduce:generate:37: rank=19 size=32
generate
889889933:0:21:EmberEngine:Reduce:generate:37: rank=21 size=32
generate
889923533:0:16:EmberEngine:Reduce:generate:37: rank=16 size=32
generate
889966733:0:25:EmberEngine:Reduce:generate:37: rank=25 size=32
generate
889976333:0:24:EmberEngine:Reduce:generate:37: rank=24 size=32
generate
890010733:0:26:EmberEngine:Reduce:generate:37: rank=26 size=32
generate
890053133:0:27:EmberEngine:Reduce:generate:37: rank=27 size=32
generate
890062733:0:20:EmberEngine:Reduce:generate:37: rank=20 size=32
generate
890097133:0:29:EmberEngine:Reduce:generate:37: rank=29 size=32
generate
890105933:0:22:EmberEngine:Reduce:generate:37: rank=22 size=32
generate
890106733:0:28:EmberEngine:Reduce:generate:37: rank=28 size=32
generate
890140333:0:30:EmberEngine:Reduce:generate:37: rank=30 size=32
generate
Bcast: ranks 32, loop 2, bytes 56, latency 1.893 us
890463533:0:31:EmberEngine:Reduce:generate:37: rank=31 size=32
Allgather: ranks 32, loop 3, bytes 12, latency 6.553 us
Barrier: ranks 32, loop 99, latency 9.614 us
1890917933:0:0:EmberEngine:PingPong:generate:65: rank=0 size=32
rank2=1 messageSize=256 iterations=35

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1891761133:0:1:EmberEngine:PingPong:generate:65: rank=1 size=32
1892910733:0:4:EmberEngine:PingPong:generate:65: rank=4 size=32
1894851533:0:16:EmberEngine:PingPong:generate:65: rank=16 size=32
1895392333:0:31:EmberEngine:PingPong:generate:65: rank=31 size=32
PingPong: otherRank 1, total time 63.474 us, loop 35, bufLen 256, latency 0.907 us. bandwidth 0.282320 G
B/s
1954391533:0:0:EmberEngine:PingPong:generate:65: rank=0 size=32
rank2=4 messageSize=256 iterations=35
PingPong: otherRank 4, total time 75.320 us, loop 35, bufLen 256, latency 1.076 us. bandwidth 0.237918 G
B/s
2029711533:0:0:EmberEngine:PingPong:generate:65: rank=0 size=32
rank2=16 messageSize=256 iterations=35
PingPong: otherRank 16, total time 75.320 us, loop 35, bufLen 256, latency 1.076 us. bandwidth 0.237918
GB/s
2105031533:0:0:EmberEngine:PingPong:generate:65: rank=0 size=32
rank2=31 messageSize=256 iterations=35
PingPong: otherRank 31, total time 99.512 us, loop 35, bufLen 256, latency 1.422 us. bandwidth 0.180079
GB/s
Simulation is complete, simulated time: 2.21284 ms
=== MiniFE mesh 8x8 (PPN=2) ===
=====
Mesh shape      : 8x8
Routers        : 64
Nodes per router : 2 (local_ports)
Total nodes     : 128
Total MPI ranks : 128
PPN (ranks/node) : 2  <-- should be 2
Stats CSV       : /Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/MiniFE/Mesh/PPN_2/results
/mesh_8x8_ppn2_nodes128_stats.csv
=====
Allreduce: ranks 128, loop 92, 1 double(s), latency 15.116 us
generate
1394128733:0:0:EmberEngine:Reduce:generate:37: rank=0 size=128
generate
1395471933:0:1:EmberEngine:Reduce:generate:37: rank=1 size=128
generate
1395687933:0:2:EmberEngine:Reduce:generate:37: rank=2 size=128
generate
1396401533:0:3:EmberEngine:Reduce:generate:37: rank=3 size=128
generate
1396617533:0:5:EmberEngine:Reduce:generate:37: rank=5 size=128
generate
1396626933:0:4:EmberEngine:Reduce:generate:37: rank=4 size=128
generate
1396834133:0:6:EmberEngine:Reduce:generate:37: rank=6 size=128
generate
1397417533:0:7:EmberEngine:Reduce:generate:37: rank=7 size=128
generate
1397642933:0:8:EmberEngine:Reduce:generate:37: rank=8 size=128
generate
1397652533:0:9:EmberEngine:Reduce:generate:37: rank=9 size=128
generate
1397719933:0:11:EmberEngine:Reduce:generate:37: rank=11 size=128
generate
1397869333:0:10:EmberEngine:Reduce:generate:37: rank=10 size=128
generate
1397936533:0:13:EmberEngine:Reduce:generate:37: rank=13 size=128
generate
1397946133:0:12:EmberEngine:Reduce:generate:37: rank=12 size=128
generate
1398153133:0:14:EmberEngine:Reduce:generate:37: rank=14 size=128
generate
1398606333:0:15:EmberEngine:Reduce:generate:37: rank=15 size=128
generate
1398736533:0:16:EmberEngine:Reduce:generate:37: rank=16 size=128
generate
1398822333:0:23:EmberEngine:Reduce:generate:37: rank=23 size=128
generate
1398841333:0:19:EmberEngine:Reduce:generate:37: rank=19 size=128
generate
1398866133:0:24:EmberEngine:Reduce:generate:37: rank=24 size=128
generate
1398885133:0:20:EmberEngine:Reduce:generate:37: rank=20 size=128
generate
1398918133:0:17:EmberEngine:Reduce:generate:37: rank=17 size=128
generate

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1398952533:0:27:EmberEngine:Reduce:generate:37: rank=27 size=128
generate
1398961933:0:18:EmberEngine:Reduce:generate:37: rank=18 size=128
generate
1398996333:0:28:EmberEngine:Reduce:generate:37: rank=28 size=128
generate
1399048533:0:25:EmberEngine:Reduce:generate:37: rank=25 size=128
generate
1399058133:0:21:EmberEngine:Reduce:generate:37: rank=21 size=128
generate
1399092333:0:26:EmberEngine:Reduce:generate:37: rank=26 size=128
generate
1399101933:0:22:EmberEngine:Reduce:generate:37: rank=22 size=128
generate
1399169133:0:29:EmberEngine:Reduce:generate:37: rank=29 size=128
generate
1399212333:0:30:EmberEngine:Reduce:generate:37: rank=30 size=128
generate
1399535933:0:31:EmberEngine:Reduce:generate:37: rank=31 size=128
generate
1399666133:0:33:EmberEngine:Reduce:generate:37: rank=33 size=128
generate
1399882733:0:34:EmberEngine:Reduce:generate:37: rank=34 size=128
generate
1399934133:0:35:EmberEngine:Reduce:generate:37: rank=35 size=128
generate
1399943733:0:39:EmberEngine:Reduce:generate:37: rank=39 size=128
generate
1399977933:0:37:EmberEngine:Reduce:generate:37: rank=37 size=128
generate
1399987533:0:41:EmberEngine:Reduce:generate:37: rank=41 size=128
generate
1400097533:0:47:EmberEngine:Reduce:generate:37: rank=47 size=128
generate
1400141333:0:55:EmberEngine:Reduce:generate:37: rank=55 size=128
generate
1400150733:0:36:EmberEngine:Reduce:generate:37: rank=36 size=128
generate
1400169933:0:40:EmberEngine:Reduce:generate:37: rank=40 size=128
generate
1400185133:0:57:EmberEngine:Reduce:generate:37: rank=57 size=128
generate
1400193933:0:38:EmberEngine:Reduce:generate:37: rank=38 size=128
generate
1400194733:0:56:EmberEngine:Reduce:generate:37: rank=56 size=128
generate
1400203533:0:42:EmberEngine:Reduce:generate:37: rank=42 size=128
generate
1400227733:0:49:EmberEngine:Reduce:generate:37: rank=49 size=128
generate
1400228333:0:58:EmberEngine:Reduce:generate:37: rank=58 size=128
generate
1400237333:0:48:EmberEngine:Reduce:generate:37: rank=48 size=128
generate
1400255733:0:43:EmberEngine:Reduce:generate:37: rank=43 size=128
generate
1400271533:0:50:EmberEngine:Reduce:generate:37: rank=50 size=128
generate
1400271533:0:59:EmberEngine:Reduce:generate:37: rank=59 size=128
generate
1400299533:0:45:EmberEngine:Reduce:generate:37: rank=45 size=128
generate
1400314733:0:61:EmberEngine:Reduce:generate:37: rank=61 size=128
generate
1400323733:0:51:EmberEngine:Reduce:generate:37: rank=51 size=128
generate
1400324333:0:60:EmberEngine:Reduce:generate:37: rank=60 size=128
generate
1400357333:0:32:EmberEngine:Reduce:generate:37: rank=32 size=128
generate
1400357933:0:62:EmberEngine:Reduce:generate:37: rank=62 size=128
generate
1400367533:0:52:EmberEngine:Reduce:generate:37: rank=52 size=128
generate
1400377133:0:53:EmberEngine:Reduce:generate:37: rank=53 size=128
generate

1400410733:0:54:EmberEngine:Reduce:generate:37: rank=54 size=128
generate
1400421533:0:63:EmberEngine:Reduce:generate:37: rank=63 size=128
generate
1400462733:0:44:EmberEngine:Reduce:generate:37: rank=44 size=128
generate
1400506733:0:46:EmberEngine:Reduce:generate:37: rank=46 size=128
generate
1400508333:0:67:EmberEngine:Reduce:generate:37: rank=67 size=128
generate
1400725133:0:69:EmberEngine:Reduce:generate:37: rank=69 size=128
generate
1400734733:0:68:EmberEngine:Reduce:generate:37: rank=68 size=128
generate
1400862733:0:71:EmberEngine:Reduce:generate:37: rank=71 size=128
generate
1400939533:0:95:EmberEngine:Reduce:generate:37: rank=95 size=128
generate
1400941133:0:70:EmberEngine:Reduce:generate:37: rank=70 size=128
generate
1400993133:0:75:EmberEngine:Reduce:generate:37: rank=75 size=128
generate
1401045133:0:79:EmberEngine:Reduce:generate:37: rank=79 size=128
generate
1401079533:0:73:EmberEngine:Reduce:generate:37: rank=73 size=128
generate
1401079533:0:97:EmberEngine:Reduce:generate:37: rank=97 size=128
generate
1401089133:0:72:EmberEngine:Reduce:generate:37: rank=72 size=128
generate
1401089133:0:83:EmberEngine:Reduce:generate:37: rank=83 size=128
generate
1401113133:0:65:EmberEngine:Reduce:generate:37: rank=65 size=128
generate
1401122733:0:64:EmberEngine:Reduce:generate:37: rank=64 size=128
generate
1401132333:0:84:EmberEngine:Reduce:generate:37: rank=84 size=128
generate
1401156333:0:99:EmberEngine:Reduce:generate:37: rank=99 size=128
generate
1401185133:0:80:EmberEngine:Reduce:generate:37: rank=80 size=128
generate
1401200333:0:101:EmberEngine:Reduce:generate:37: rank=101 size=128
generate
1401209133:0:77:EmberEngine:Reduce:generate:37: rank=77 size=128
generate
1401218733:0:76:EmberEngine:Reduce:generate:37: rank=76 size=128
generate
1401228333:0:91:EmberEngine:Reduce:generate:37: rank=91 size=128
generate
1401270733:0:87:EmberEngine:Reduce:generate:37: rank=87 size=128
generate
1401271533:0:92:EmberEngine:Reduce:generate:37: rank=92 size=128
generate
1401295533:0:74:EmberEngine:Reduce:generate:37: rank=74 size=128
generate
1401296333:0:98:EmberEngine:Reduce:generate:37: rank=98 size=128
generate
1401305133:0:85:EmberEngine:Reduce:generate:37: rank=85 size=128
generate
1401314733:0:88:EmberEngine:Reduce:generate:37: rank=88 size=128
generate
1401329133:0:111:EmberEngine:Reduce:generate:37: rank=111 size=128
generate
1401329933:0:66:EmberEngine:Reduce:generate:37: rank=66 size=128
generate
1401329933:0:123:EmberEngine:Reduce:generate:37: rank=123 size=128
generate
1401338733:0:103:EmberEngine:Reduce:generate:37: rank=103 size=128
generate
1401348333:0:86:EmberEngine:Reduce:generate:37: rank=86 size=128
generate
1401357933:0:81:EmberEngine:Reduce:generate:37: rank=81 size=128
generate
1401373133:0:100:EmberEngine:Reduce:generate:37: rank=100 size=128
generate


```

1401373133:0:115:EmberEngine:Reduce:generate:37: rank=115 size=128
generate
1401373133:0:119:EmberEngine:Reduce:generate:37: rank=119 size=128
generate
1401373133:0:125:EmberEngine:Reduce:generate:37: rank=125 size=128
generate
1401382733:0:105:EmberEngine:Reduce:generate:37: rank=105 size=128
generate
1401382733:0:124:EmberEngine:Reduce:generate:37: rank=124 size=128
generate
1401401133:0:82:EmberEngine:Reduce:generate:37: rank=82 size=128
generate
1401416333:0:102:EmberEngine:Reduce:generate:37: rank=102 size=128
generate
1401416333:0:117:EmberEngine:Reduce:generate:37: rank=117 size=128
generate
1401416333:0:120:EmberEngine:Reduce:generate:37: rank=120 size=128
generate
1401425133:0:78:EmberEngine:Reduce:generate:37: rank=78 size=128
generate
1401425133:0:126:EmberEngine:Reduce:generate:37: rank=126 size=128
generate
1401425933:0:116:EmberEngine:Reduce:generate:37: rank=116 size=128
generate
1401425933:0:121:EmberEngine:Reduce:generate:37: rank=121 size=128
generate
1401435533:0:93:EmberEngine:Reduce:generate:37: rank=93 size=128
generate
1401459533:0:118:EmberEngine:Reduce:generate:37: rank=118 size=128
generate
1401469133:0:112:EmberEngine:Reduce:generate:37: rank=112 size=128
generate
1401469133:0:122:EmberEngine:Reduce:generate:37: rank=122 size=128
generate
1401477933:0:107:EmberEngine:Reduce:generate:37: rank=107 size=128
generate
1401478733:0:89:EmberEngine:Reduce:generate:37: rank=89 size=128
generate
1401478733:0:94:EmberEngine:Reduce:generate:37: rank=94 size=128
generate
1401478733:0:113:EmberEngine:Reduce:generate:37: rank=113 size=128
generate
Bcast: ranks 128, loop 2, bytes 56, latency 1.893 us
1401480333:0:127:EmberEngine:Reduce:generate:37: rank=127 size=128
generate
1401512333:0:109:EmberEngine:Reduce:generate:37: rank=109 size=128
generate
1401521933:0:90:EmberEngine:Reduce:generate:37: rank=90 size=128
generate
1401521933:0:114:EmberEngine:Reduce:generate:37: rank=114 size=128
generate
1401555533:0:104:EmberEngine:Reduce:generate:37: rank=104 size=128
generate
1401598733:0:106:EmberEngine:Reduce:generate:37: rank=106 size=128
generate
1401694733:0:108:EmberEngine:Reduce:generate:37: rank=108 size=128
generate
1401728333:0:110:EmberEngine:Reduce:generate:37: rank=110 size=128
generate
1401761133:0:96:EmberEngine:Reduce:generate:37: rank=96 size=128
Allgather: ranks 128, loop 3, bytes 12, latency 9.152 us
Barrier: ranks 128, loop 99, latency 15.163 us
2962776333:0:0:EmberEngine:PingPong:generate:65: rank=0 size=128
rank2=1 messageSize=256 iterations=35
2963619533:0:1:EmberEngine:PingPong:generate:65: rank=1 size=128
2965785133:0:8:EmberEngine:PingPong:generate:65: rank=8 size=128
2969264333:0:64:EmberEngine:PingPong:generate:65: rank=64 size=128
2969628333:0:127:EmberEngine:PingPong:generate:65: rank=127 size=128
PingPong: otherRank 1, total time 63.473 us, loop 35, bufLen 256, latency 0.907 us. bandwidth 0.282325 G
B/s
3026249933:0:0:EmberEngine:PingPong:generate:65: rank=0 size=128
rank2=8 messageSize=256 iterations=35
PingPong: otherRank 8, total time 87.416 us, loop 35, bufLen 256, latency 1.249 us. bandwidth 0.204997 G
B/s
3113665933:0:0:EmberEngine:PingPong:generate:65: rank=0 size=128
rank2=64 messageSize=256 iterations=35

```

```

PingPong: otherRank 64, total time 87.416 us, loop 35, bufLen 256, latency 1.249 us. bandwidth 0.204997
GB/s
3201081933:0:0:EmberEngine:PingPong:generate:65: rank=0 size=128
rank2=127 messageSize=256 iterations=35
PingPong: otherRank 127, total time 147.896 us, loop 35, bufLen 256, latency 2.113 us. bandwidth 0.12116
6 GB/s
Simulation is complete, simulated time: 3.36162 ms
=== MiniFE mesh 16x16 (PPN=2) ===
=====
Mesh shape      : 16x16
Routers         : 256
Nodes per router : 2 (local_ports)
Total nodes     : 512
Total MPI ranks  : 512
PPN (ranks/node) : 2  <-- should be 2
Stats CSV       : /Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/MiniFE/Mesh/PPN_2/results
/mesh_16x16_ppn2_nodes512_stats.csv
=====
Allreduce: ranks 512, loop 92, 1 double(s), latency 23.566 us
generate
2171573533:0:0:EmberEngine:Reduce:generate:37: rank=0 size=512
generate
2172916733:0:1:EmberEngine:Reduce:generate:37: rank=1 size=512
generate
2173132733:0:2:EmberEngine:Reduce:generate:37: rank=2 size=512
generate
2173846333:0:3:EmberEngine:Reduce:generate:37: rank=3 size=512
generate
2174062333:0:5:EmberEngine:Reduce:generate:37: rank=5 size=512
generate
2174071733:0:4:EmberEngine:Reduce:generate:37: rank=4 size=512
generate
2174278933:0:6:EmberEngine:Reduce:generate:37: rank=6 size=512
generate
2174862333:0:7:EmberEngine:Reduce:generate:37: rank=7 size=512
generate
2175087733:0:8:EmberEngine:Reduce:generate:37: rank=8 size=512
generate
2175097333:0:9:EmberEngine:Reduce:generate:37: rank=9 size=512
generate
2175164733:0:11:EmberEngine:Reduce:generate:37: rank=11 size=512
generate
2175314133:0:10:EmberEngine:Reduce:generate:37: rank=10 size=512
generate
2175381333:0:13:EmberEngine:Reduce:generate:37: rank=13 size=512
generate
2175390933:0:12:EmberEngine:Reduce:generate:37: rank=12 size=512
generate
2175597933:0:14:EmberEngine:Reduce:generate:37: rank=14 size=512
generate
2176051133:0:15:EmberEngine:Reduce:generate:37: rank=15 size=512
generate
2176276533:0:16:EmberEngine:Reduce:generate:37: rank=16 size=512
generate
2176286133:0:17:EmberEngine:Reduce:generate:37: rank=17 size=512
generate
2176381333:0:19:EmberEngine:Reduce:generate:37: rank=19 size=512
generate
2176511733:0:18:EmberEngine:Reduce:generate:37: rank=18 size=512
generate
2176526333:0:23:EmberEngine:Reduce:generate:37: rank=23 size=512
generate
2176617133:0:20:EmberEngine:Reduce:generate:37: rank=20 size=512
generate
2176617333:0:21:EmberEngine:Reduce:generate:37: rank=21 size=512
generate
2176752533:0:24:EmberEngine:Reduce:generate:37: rank=24 size=512
generate
2176762133:0:25:EmberEngine:Reduce:generate:37: rank=25 size=512
generate
2176814733:0:22:EmberEngine:Reduce:generate:37: rank=22 size=512
generate
2176829333:0:27:EmberEngine:Reduce:generate:37: rank=27 size=512
generate
2176978733:0:26:EmberEngine:Reduce:generate:37: rank=26 size=512
generate

```

2177045933:0:29:EmberEngine:Reduce:generate:37: rank=29 size=512
generate
2177055533:0:28:EmberEngine:Reduce:generate:37: rank=28 size=512
generate
2177261933:0:30:EmberEngine:Reduce:generate:37: rank=30 size=512
generate
2177585533:0:31:EmberEngine:Reduce:generate:37: rank=31 size=512
generate
2177715733:0:32:EmberEngine:Reduce:generate:37: rank=32 size=512
generate
2177806133:0:47:EmberEngine:Reduce:generate:37: rank=47 size=512
generate
2177820533:0:35:EmberEngine:Reduce:generate:37: rank=35 size=512
generate
2177829333:0:39:EmberEngine:Reduce:generate:37: rank=39 size=512
generate
2177849333:0:48:EmberEngine:Reduce:generate:37: rank=48 size=512
generate
2177864333:0:36:EmberEngine:Reduce:generate:37: rank=36 size=512
generate
2177882733:0:40:EmberEngine:Reduce:generate:37: rank=40 size=512
generate
2177897333:0:33:EmberEngine:Reduce:generate:37: rank=33 size=512
generate
2177931733:0:55:EmberEngine:Reduce:generate:37: rank=55 size=512
generate
2177941133:0:34:EmberEngine:Reduce:generate:37: rank=34 size=512
generate
2177950933:0:51:EmberEngine:Reduce:generate:37: rank=51 size=512
generate
2177975533:0:56:EmberEngine:Reduce:generate:37: rank=56 size=512
generate
2177993133:0:43:EmberEngine:Reduce:generate:37: rank=43 size=512
generate
2177994733:0:52:EmberEngine:Reduce:generate:37: rank=52 size=512
generate
2178027733:0:49:EmberEngine:Reduce:generate:37: rank=49 size=512
generate
2178037133:0:44:EmberEngine:Reduce:generate:37: rank=44 size=512
generate
2178054933:0:37:EmberEngine:Reduce:generate:37: rank=37 size=512
generate
2178061933:0:59:EmberEngine:Reduce:generate:37: rank=59 size=512
generate
2178069933:0:41:EmberEngine:Reduce:generate:37: rank=41 size=512
generate
2178071533:0:50:EmberEngine:Reduce:generate:37: rank=50 size=512
generate
2178089933:0:38:EmberEngine:Reduce:generate:37: rank=38 size=512
generate
2178105133:0:60:EmberEngine:Reduce:generate:37: rank=60 size=512
generate
2178113933:0:42:EmberEngine:Reduce:generate:37: rank=42 size=512
generate
2178157933:0:57:EmberEngine:Reduce:generate:37: rank=57 size=512
generate
2178167533:0:53:EmberEngine:Reduce:generate:37: rank=53 size=512
generate
2178176333:0:45:EmberEngine:Reduce:generate:37: rank=45 size=512
generate
2178201133:0:58:EmberEngine:Reduce:generate:37: rank=58 size=512
generate
2178210733:0:54:EmberEngine:Reduce:generate:37: rank=54 size=512
generate
2178219533:0:46:EmberEngine:Reduce:generate:37: rank=46 size=512
generate
2178277933:0:61:EmberEngine:Reduce:generate:37: rank=61 size=512
generate
2178321133:0:62:EmberEngine:Reduce:generate:37: rank=62 size=512
generate
2178515133:0:63:EmberEngine:Reduce:generate:37: rank=63 size=512
generate
2178645333:0:65:EmberEngine:Reduce:generate:37: rank=65 size=512
generate
2178861933:0:66:EmberEngine:Reduce:generate:37: rank=66 size=512
generate

2178913333:0:67:EmberEngine:Reduce:generate:37: rank=67 size=512
generate
2178922933:0:71:EmberEngine:Reduce:generate:37: rank=71 size=512
generate
2178957133:0:69:EmberEngine:Reduce:generate:37: rank=69 size=512
generate
2178966733:0:73:EmberEngine:Reduce:generate:37: rank=73 size=512
generate
2179105133:0:79:EmberEngine:Reduce:generate:37: rank=79 size=512
generate
2179129933:0:68:EmberEngine:Reduce:generate:37: rank=68 size=512
generate
2179139533:0:72:EmberEngine:Reduce:generate:37: rank=72 size=512
generate
2179157933:0:81:EmberEngine:Reduce:generate:37: rank=81 size=512
generate
2179173133:0:70:EmberEngine:Reduce:generate:37: rank=70 size=512
generate
2179182733:0:74:EmberEngine:Reduce:generate:37: rank=74 size=512
generate
2179243733:0:75:EmberEngine:Reduce:generate:37: rank=75 size=512
generate
2179250733:0:119:EmberEngine:Reduce:generate:37: rank=119 size=512
generate
2179278733:0:77:EmberEngine:Reduce:generate:37: rank=77 size=512
generate
2179293933:0:121:EmberEngine:Reduce:generate:37: rank=121 size=512
generate
2179297933:0:111:EmberEngine:Reduce:generate:37: rank=111 size=512
generate
2179303533:0:120:EmberEngine:Reduce:generate:37: rank=120 size=512
generate
2179321933:0:80:EmberEngine:Reduce:generate:37: rank=80 size=512
generate
2179337133:0:113:EmberEngine:Reduce:generate:37: rank=113 size=512
generate
2179337133:0:122:EmberEngine:Reduce:generate:37: rank=122 size=512
generate
2179346733:0:112:EmberEngine:Reduce:generate:37: rank=112 size=512
generate
2179374733:0:82:EmberEngine:Reduce:generate:37: rank=82 size=512
generate
2179380333:0:114:EmberEngine:Reduce:generate:37: rank=114 size=512
generate
2179380333:0:123:EmberEngine:Reduce:generate:37: rank=123 size=512
generate
2179423533:0:125:EmberEngine:Reduce:generate:37: rank=125 size=512
generate
2179426933:0:95:EmberEngine:Reduce:generate:37: rank=95 size=512
generate
2179431533:0:83:EmberEngine:Reduce:generate:37: rank=83 size=512
generate
2179433133:0:115:EmberEngine:Reduce:generate:37: rank=115 size=512
generate
2179433133:0:124:EmberEngine:Reduce:generate:37: rank=124 size=512
generate
2179442733:0:87:EmberEngine:Reduce:generate:37: rank=87 size=512
generate
2179460333:0:76:EmberEngine:Reduce:generate:37: rank=76 size=512
generate
2179466733:0:126:EmberEngine:Reduce:generate:37: rank=126 size=512
generate
2179475533:0:85:EmberEngine:Reduce:generate:37: rank=85 size=512
generate
2179476333:0:116:EmberEngine:Reduce:generate:37: rank=116 size=512
generate
2179485133:0:89:EmberEngine:Reduce:generate:37: rank=89 size=512
generate
2179485333:0:103:EmberEngine:Reduce:generate:37: rank=103 size=512
generate
2179485933:0:117:EmberEngine:Reduce:generate:37: rank=117 size=512
generate
2179494733:0:78:EmberEngine:Reduce:generate:37: rank=78 size=512
generate
2179519533:0:118:EmberEngine:Reduce:generate:37: rank=118 size=512
generate

2179531133:0:127:EmberEngine:Reduce:generate:37: rank=127 size=512
generate
2179538733:0:105:EmberEngine:Reduce:generate:37: rank=105 size=512
generate
2179548333:0:104:EmberEngine:Reduce:generate:37: rank=104 size=512
generate
2179556533:0:97:EmberEngine:Reduce:generate:37: rank=97 size=512
generate
2179566133:0:96:EmberEngine:Reduce:generate:37: rank=96 size=512
generate
2179572333:0:106:EmberEngine:Reduce:generate:37: rank=106 size=512
generate
2179600333:0:98:EmberEngine:Reduce:generate:37: rank=98 size=512
generate
2179615533:0:107:EmberEngine:Reduce:generate:37: rank=107 size=512
generate
2179647533:0:84:EmberEngine:Reduce:generate:37: rank=84 size=512
generate
2179648533:0:99:EmberEngine:Reduce:generate:37: rank=99 size=512
generate
2179657933:0:88:EmberEngine:Reduce:generate:37: rank=88 size=512
generate
2179658733:0:108:EmberEngine:Reduce:generate:37: rank=108 size=512
generate
2179668333:0:109:EmberEngine:Reduce:generate:37: rank=109 size=512
generate
2179691533:0:86:EmberEngine:Reduce:generate:37: rank=86 size=512
generate
2179692333:0:100:EmberEngine:Reduce:generate:37: rank=100 size=512
generate
2179701133:0:90:EmberEngine:Reduce:generate:37: rank=90 size=512
generate
2179701933:0:101:EmberEngine:Reduce:generate:37: rank=101 size=512
generate
2179701933:0:110:EmberEngine:Reduce:generate:37: rank=110 size=512
generate
2179718933:0:91:EmberEngine:Reduce:generate:37: rank=91 size=512
generate
2179735533:0:102:EmberEngine:Reduce:generate:37: rank=102 size=512
generate
2179747733:0:131:EmberEngine:Reduce:generate:37: rank=131 size=512
generate
2179753933:0:93:EmberEngine:Reduce:generate:37: rank=93 size=512
generate
2179926733:0:92:EmberEngine:Reduce:generate:37: rank=92 size=512
generate
2179964333:0:133:EmberEngine:Reduce:generate:37: rank=133 size=512
generate
2179969933:0:94:EmberEngine:Reduce:generate:37: rank=94 size=512
generate
2179973933:0:132:EmberEngine:Reduce:generate:37: rank=132 size=512
generate
2180027733:0:64:EmberEngine:Reduce:generate:37: rank=64 size=512
generate
2180102133:0:135:EmberEngine:Reduce:generate:37: rank=135 size=512
generate
2180180333:0:134:EmberEngine:Reduce:generate:37: rank=134 size=512
generate
2180232333:0:139:EmberEngine:Reduce:generate:37: rank=139 size=512
generate
2180284533:0:143:EmberEngine:Reduce:generate:37: rank=143 size=512
generate
2180318733:0:137:EmberEngine:Reduce:generate:37: rank=137 size=512
generate
2180328333:0:136:EmberEngine:Reduce:generate:37: rank=136 size=512
generate
2180414733:0:147:EmberEngine:Reduce:generate:37: rank=147 size=512
generate
2180448333:0:141:EmberEngine:Reduce:generate:37: rank=141 size=512
generate
2180467533:0:140:EmberEngine:Reduce:generate:37: rank=140 size=512
generate
2180501133:0:145:EmberEngine:Reduce:generate:37: rank=145 size=512
generate
2180510733:0:144:EmberEngine:Reduce:generate:37: rank=144 size=512
generate

2180530733:0:191:EmberEngine:Reduce:generate:37: rank=191 size=512
generate
2180534733:0:138:EmberEngine:Reduce:generate:37: rank=138 size=512
generate
2180589533:0:255:EmberEngine:Reduce:generate:37: rank=255 size=512
generate
2180630733:0:149:EmberEngine:Reduce:generate:37: rank=149 size=512
generate
2180649933:0:148:EmberEngine:Reduce:generate:37: rank=148 size=512
generate
2180664333:0:142:EmberEngine:Reduce:generate:37: rank=142 size=512
generate
2180674733:0:193:EmberEngine:Reduce:generate:37: rank=193 size=512
generate
2180698733:0:251:EmberEngine:Reduce:generate:37: rank=251 size=512
generate
2180717133:0:146:EmberEngine:Reduce:generate:37: rank=146 size=512
generate
2180741933:0:243:EmberEngine:Reduce:generate:37: rank=243 size=512
generate
2180741933:0:247:EmberEngine:Reduce:generate:37: rank=247 size=512
generate
2180744533:0:253:EmberEngine:Reduce:generate:37: rank=253 size=512
generate
2180751533:0:195:EmberEngine:Reduce:generate:37: rank=195 size=512
generate
2180761133:0:252:EmberEngine:Reduce:generate:37: rank=252 size=512
generate
2180779533:0:151:EmberEngine:Reduce:generate:37: rank=151 size=512
generate
2180785133:0:239:EmberEngine:Reduce:generate:37: rank=239 size=512
generate
2180785133:0:245:EmberEngine:Reduce:generate:37: rank=245 size=512
generate
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generate
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generate

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generate
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generate
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generate
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generate
2181965933:0:321:EmberEngine:Reduce:generate:37: rank=321 size=512
generate
2181972333:0:256:EmberEngine:Reduce:generate:37: rank=256 size=512
generate
2181981933:0:257:EmberEngine:Reduce:generate:37: rank=257 size=512
generate
2181981933:0:283:EmberEngine:Reduce:generate:37: rank=283 size=512
generate
Bcast: ranks 512, loop 2, bytes 56, latency 1.893 us
2181993933:0:511:EmberEngine:Reduce:generate:37: rank=511 size=512
generate
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generate
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generate
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generate
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generate
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generate
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generate
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generate
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generate
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generate
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generate
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generate
2182121133:0:291:EmberEngine:Reduce:generate:37: rank=291 size=512
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generate
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generate
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generate
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generate
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generate
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generate
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generate
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generate
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generate
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generate
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generate
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generate
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generate
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2182553933:0:362:EmberEngine:Reduce:generate:37: rank=362 size=512
generate
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generate
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generate
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generate
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generate
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generate
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generate
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2182600333:0:401:EmberEngine:Reduce:generate:37: rank=401 size=512
generate
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generate
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generate
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generate
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generate
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generate
2182615533:0:372:EmberEngine:Reduce:generate:37: rank=372 size=512
generate
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generate
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generate
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generate
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generate
2182645933:0:490:EmberEngine:Reduce:generate:37: rank=490 size=512
generate
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generate
2182657933:0:365:EmberEngine:Reduce:generate:37: rank=365 size=512
generate
2182664333:0:479:EmberEngine:Reduce:generate:37: rank=479 size=512
generate
2182667533:0:461:EmberEngine:Reduce:generate:37: rank=461 size=512
generate
2182673933:0:463:EmberEngine:Reduce:generate:37: rank=463 size=512
generate
2182682733:0:366:EmberEngine:Reduce:generate:37: rank=366 size=512
generate
2182683533:0:407:EmberEngine:Reduce:generate:37: rank=407 size=512
generate
2182683533:0:443:EmberEngine:Reduce:generate:37: rank=443 size=512
generate
2182686733:0:338:EmberEngine:Reduce:generate:37: rank=338 size=512
generate
2182707533:0:456:EmberEngine:Reduce:generate:37: rank=456 size=512
generate
2182717133:0:465:EmberEngine:Reduce:generate:37: rank=465 size=512
generate
2182720333:0:339:EmberEngine:Reduce:generate:37: rank=339 size=512
generate
2182721133:0:405:EmberEngine:Reduce:generate:37: rank=405 size=512
generate
2182725933:0:415:EmberEngine:Reduce:generate:37: rank=415 size=512
generate
2182726733:0:435:EmberEngine:Reduce:generate:37: rank=435 size=512
generate
2182726733:0:444:EmberEngine:Reduce:generate:37: rank=444 size=512
generate
2182727533:0:452:EmberEngine:Reduce:generate:37: rank=452 size=512
generate
2182734733:0:352:EmberEngine:Reduce:generate:37: rank=352 size=512
generate
2182736333:0:427:EmberEngine:Reduce:generate:37: rank=427 size=512
generate
2182744333:0:353:EmberEngine:Reduce:generate:37: rank=353 size=512
generate
2182753933:0:404:EmberEngine:Reduce:generate:37: rank=404 size=512
generate
2182754733:0:439:EmberEngine:Reduce:generate:37: rank=439 size=512
generate
2182754733:0:483:EmberEngine:Reduce:generate:37: rank=483 size=512
generate
2182760333:0:458:EmberEngine:Reduce:generate:37: rank=458 size=512
generate
2182763533:0:341:EmberEngine:Reduce:generate:37: rank=341 size=512
generate
2182769933:0:431:EmberEngine:Reduce:generate:37: rank=431 size=512
generate
2182769933:0:436:EmberEngine:Reduce:generate:37: rank=436 size=512
generate
2182769933:0:454:EmberEngine:Reduce:generate:37: rank=454 size=512
generate
2182777133:0:344:EmberEngine:Reduce:generate:37: rank=344 size=512
generate
2182779533:0:423:EmberEngine:Reduce:generate:37: rank=423 size=512
generate
2182779533:0:428:EmberEngine:Reduce:generate:37: rank=428 size=512
generate
2182787533:0:354:EmberEngine:Reduce:generate:37: rank=354 size=512
generate
2182789133:0:419:EmberEngine:Reduce:generate:37: rank=419 size=512
generate
2182794733:0:485:EmberEngine:Reduce:generate:37: rank=485 size=512
generate
2182797133:0:440:EmberEngine:Reduce:generate:37: rank=440 size=512

generate
2182803533:0:411:EmberEngine:Reduce:generate:37: rank=411 size=512
generate
2182803533:0:480:EmberEngine:Reduce:generate:37: rank=480 size=512
generate
2182804333:0:484:EmberEngine:Reduce:generate:37: rank=484 size=512
generate
2182807533:0:402:EmberEngine:Reduce:generate:37: rank=402 size=512
generate
2182812333:0:346:EmberEngine:Reduce:generate:37: rank=346 size=512
generate
2182812333:0:355:EmberEngine:Reduce:generate:37: rank=355 size=512
generate
2182822733:0:432:EmberEngine:Reduce:generate:37: rank=432 size=512
generate
2182822733:0:467:EmberEngine:Reduce:generate:37: rank=467 size=512
generate
2182822733:0:481:EmberEngine:Reduce:generate:37: rank=481 size=512
generate
2182831533:0:471:EmberEngine:Reduce:generate:37: rank=471 size=512
generate
2182831533:0:486:EmberEngine:Reduce:generate:37: rank=486 size=512
generate
2182832333:0:420:EmberEngine:Reduce:generate:37: rank=420 size=512
generate
2182832333:0:424:EmberEngine:Reduce:generate:37: rank=424 size=512
generate
2182837133:0:460:EmberEngine:Reduce:generate:37: rank=460 size=512
generate
2182846733:0:469:EmberEngine:Reduce:generate:37: rank=469 size=512
generate
2182853933:0:347:EmberEngine:Reduce:generate:37: rank=347 size=512
generate
2182855533:0:356:EmberEngine:Reduce:generate:37: rank=356 size=512
generate
2182856333:0:416:EmberEngine:Reduce:generate:37: rank=416 size=512
generate
2182865933:0:473:EmberEngine:Reduce:generate:37: rank=473 size=512
generate
2182871533:0:482:EmberEngine:Reduce:generate:37: rank=482 size=512
generate
2182874733:0:357:EmberEngine:Reduce:generate:37: rank=357 size=512
generate
2182884333:0:462:EmberEngine:Reduce:generate:37: rank=462 size=512
generate
2182889133:0:349:EmberEngine:Reduce:generate:37: rank=349 size=512
generate
2182889933:0:464:EmberEngine:Reduce:generate:37: rank=464 size=512
generate
2182899533:0:445:EmberEngine:Reduce:generate:37: rank=445 size=512
generate
2182903533:0:409:EmberEngine:Reduce:generate:37: rank=409 size=512
generate
2182907533:0:358:EmberEngine:Reduce:generate:37: rank=358 size=512
generate
2182913133:0:408:EmberEngine:Reduce:generate:37: rank=408 size=512
generate
2182931533:0:340:EmberEngine:Reduce:generate:37: rank=340 size=512
generate
2182936333:0:466:EmberEngine:Reduce:generate:37: rank=466 size=512
generate
2182937133:0:406:EmberEngine:Reduce:generate:37: rank=406 size=512
generate
2182942733:0:446:EmberEngine:Reduce:generate:37: rank=446 size=512
generate
2182942733:0:475:EmberEngine:Reduce:generate:37: rank=475 size=512
generate
2182955533:0:437:EmberEngine:Reduce:generate:37: rank=437 size=512
generate
2182961933:0:429:EmberEngine:Reduce:generate:37: rank=429 size=512
generate
2182970733:0:441:EmberEngine:Reduce:generate:37: rank=441 size=512
generate
2182975533:0:342:EmberEngine:Reduce:generate:37: rank=342 size=512
generate
2182976333:0:477:EmberEngine:Reduce:generate:37: rank=477 size=512

generate
2182985933:0:433:EmberEngine:Reduce:generate:37: rank=433 size=512
generate
2182985933:0:438:EmberEngine:Reduce:generate:37: rank=438 size=512
generate
2182995533:0:425:EmberEngine:Reduce:generate:37: rank=425 size=512
generate
2182995533:0:430:EmberEngine:Reduce:generate:37: rank=430 size=512
generate
2183005133:0:442:EmberEngine:Reduce:generate:37: rank=442 size=512
generate
2183014733:0:421:EmberEngine:Reduce:generate:37: rank=421 size=512
generate
2183019533:0:413:EmberEngine:Reduce:generate:37: rank=413 size=512
generate
2183029133:0:412:EmberEngine:Reduce:generate:37: rank=412 size=512
generate
2183029133:0:417:EmberEngine:Reduce:generate:37: rank=417 size=512
generate
2183029133:0:434:EmberEngine:Reduce:generate:37: rank=434 size=512
generate
2183032333:0:468:EmberEngine:Reduce:generate:37: rank=468 size=512
generate
2183038733:0:426:EmberEngine:Reduce:generate:37: rank=426 size=512
generate
2183047533:0:472:EmberEngine:Reduce:generate:37: rank=472 size=512
generate
2183048333:0:422:EmberEngine:Reduce:generate:37: rank=422 size=512
generate
2183062733:0:470:EmberEngine:Reduce:generate:37: rank=470 size=512
generate
2183069933:0:348:EmberEngine:Reduce:generate:37: rank=348 size=512
generate
2183072333:0:418:EmberEngine:Reduce:generate:37: rank=418 size=512
generate
2183081933:0:474:EmberEngine:Reduce:generate:37: rank=474 size=512
generate
2183105133:0:350:EmberEngine:Reduce:generate:37: rank=350 size=512
generate
2183119533:0:410:EmberEngine:Reduce:generate:37: rank=410 size=512
generate
2183146733:0:385:EmberEngine:Reduce:generate:37: rank=385 size=512
generate
2183156333:0:384:EmberEngine:Reduce:generate:37: rank=384 size=512
generate
2183158733:0:476:EmberEngine:Reduce:generate:37: rank=476 size=512
generate
2183192333:0:478:EmberEngine:Reduce:generate:37: rank=478 size=512
generate
2183235533:0:414:EmberEngine:Reduce:generate:37: rank=414 size=512
generate
2183339533:0:320:EmberEngine:Reduce:generate:37: rank=320 size=512
generate
2183362733:0:386:EmberEngine:Reduce:generate:37: rank=386 size=512
generate
2183800333:0:448:EmberEngine:Reduce:generate:37: rank=448 size=512
Allgather: ranks 512, loop 3, bytes 12, latency 14.944 us
Barrier: ranks 512, loop 99, latency 23.642 us
4606059533:0:0:EmberEngine:PingPong:generate:65: rank=0 size=512
rank2=1 messageSize=256 iterations=35
4606902733:0:1:EmberEngine:PingPong:generate:65: rank=1 size=512
4610257133:0:16:EmberEngine:PingPong:generate:65: rank=16 size=512
4615957933:0:256:EmberEngine:PingPong:generate:65: rank=256 size=512
4615980333:0:511:EmberEngine:PingPong:generate:65: rank=511 size=512
PingPong: otherRank 1, total time 63.474 us, loop 35, bufLen 256, latency 0.907 us. bandwidth 0.282320 G
B/s
4669533133:0:0:EmberEngine:PingPong:generate:65: rank=0 size=512
rank2=16 messageSize=256 iterations=35
PingPong: otherRank 16, total time 111.608 us, loop 35, bufLen 256, latency 1.594 us. bandwidth 0.160562
GB/s
4781141133:0:0:EmberEngine:PingPong:generate:65: rank=0 size=512
rank2=256 messageSize=256 iterations=35
PingPong: otherRank 256, total time 111.608 us, loop 35, bufLen 256, latency 1.594 us. bandwidth 0.16056
2 GB/s
4892749133:0:0:EmberEngine:PingPong:generate:65: rank=0 size=512
rank2=511 messageSize=256 iterations=35

PingPong: otherRank 511, total time 244.664 us, loop 35, bufLen 256, latency 3.495 us. bandwidth 0.07324 3 GB/s
Simulation is complete, simulated time: 5.15632 ms
=== MiniFE mesh 32x32 (PPN=2) ===
=====

Mesh shape	: 32x32
Routers	: 1024
Nodes per router	: 2 (local_ports)
Total nodes	: 2048
Total MPI ranks	: 2048
PPN (ranks/node)	: 2 <-- should be 2
Stats CSV	: /Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/MiniFE/Mesh/PPN_2/results /mesh_32x32_ppn2_nodes2048_stats.csv

=====

Allreduce: ranks 2048, loop 92, 1 double(s), latency 38.625 us
generate
3556972733:0:0:EmberEngine:Reduce:generate:37: rank=0 size=2048
generate
3558315933:0:1:EmberEngine:Reduce:generate:37: rank=1 size=2048
generate
3558531933:0:2:EmberEngine:Reduce:generate:37: rank=2 size=2048
generate
3559245533:0:3:EmberEngine:Reduce:generate:37: rank=3 size=2048
generate
3559461533:0:5:EmberEngine:Reduce:generate:37: rank=5 size=2048
generate
3559470933:0:4:EmberEngine:Reduce:generate:37: rank=4 size=2048
generate
3559678133:0:6:EmberEngine:Reduce:generate:37: rank=6 size=2048
generate
3560261533:0:7:EmberEngine:Reduce:generate:37: rank=7 size=2048
generate
3560486933:0:8:EmberEngine:Reduce:generate:37: rank=8 size=2048
generate
3560496533:0:9:EmberEngine:Reduce:generate:37: rank=9 size=2048
generate
3560563933:0:11:EmberEngine:Reduce:generate:37: rank=11 size=2048
generate
3560713333:0:10:EmberEngine:Reduce:generate:37: rank=10 size=2048
generate
3560780533:0:13:EmberEngine:Reduce:generate:37: rank=13 size=2048
generate
3560790133:0:12:EmberEngine:Reduce:generate:37: rank=12 size=2048
generate
3560997133:0:14:EmberEngine:Reduce:generate:37: rank=14 size=2048
generate
3561450333:0:15:EmberEngine:Reduce:generate:37: rank=15 size=2048
generate
3561675733:0:16:EmberEngine:Reduce:generate:37: rank=16 size=2048
generate
3561685333:0:17:EmberEngine:Reduce:generate:37: rank=17 size=2048
generate
3561780533:0:19:EmberEngine:Reduce:generate:37: rank=19 size=2048
generate
3561910933:0:18:EmberEngine:Reduce:generate:37: rank=18 size=2048
generate
3561925533:0:23:EmberEngine:Reduce:generate:37: rank=23 size=2048
generate
3562016333:0:20:EmberEngine:Reduce:generate:37: rank=20 size=2048
generate
3562016533:0:21:EmberEngine:Reduce:generate:37: rank=21 size=2048
generate
3562151733:0:24:EmberEngine:Reduce:generate:37: rank=24 size=2048
generate
3562161333:0:25:EmberEngine:Reduce:generate:37: rank=25 size=2048
generate
3562213933:0:22:EmberEngine:Reduce:generate:37: rank=22 size=2048
generate
3562228533:0:27:EmberEngine:Reduce:generate:37: rank=27 size=2048
generate
3562377933:0:26:EmberEngine:Reduce:generate:37: rank=26 size=2048
generate
3562445133:0:29:EmberEngine:Reduce:generate:37: rank=29 size=2048
generate
3562454733:0:28:EmberEngine:Reduce:generate:37: rank=28 size=2048
generate

3562661133:0:30:EmberEngine:Reduce:generate:37: rank=30 size=2048
generate
3562984733:0:31:EmberEngine:Reduce:generate:37: rank=31 size=2048
generate
3563210133:0:32:EmberEngine:Reduce:generate:37: rank=32 size=2048
generate
3563219733:0:33:EmberEngine:Reduce:generate:37: rank=33 size=2048
generate
3563314933:0:35:EmberEngine:Reduce:generate:37: rank=35 size=2048
generate
3563445333:0:34:EmberEngine:Reduce:generate:37: rank=34 size=2048
generate
3563497333:0:39:EmberEngine:Reduce:generate:37: rank=39 size=2048
generate
3563550933:0:36:EmberEngine:Reduce:generate:37: rank=36 size=2048
generate
3563569333:0:37:EmberEngine:Reduce:generate:37: rank=37 size=2048
generate
3563741933:0:40:EmberEngine:Reduce:generate:37: rank=40 size=2048
generate
3563751733:0:41:EmberEngine:Reduce:generate:37: rank=41 size=2048
generate
3563785933:0:38:EmberEngine:Reduce:generate:37: rank=38 size=2048
generate
3563805533:0:47:EmberEngine:Reduce:generate:37: rank=47 size=2048
generate
3563847733:0:43:EmberEngine:Reduce:generate:37: rank=43 size=2048
generate
3563968333:0:42:EmberEngine:Reduce:generate:37: rank=42 size=2048
generate
3564031733:0:48:EmberEngine:Reduce:generate:37: rank=48 size=2048
generate
3564035533:0:45:EmberEngine:Reduce:generate:37: rank=45 size=2048
generate
3564041333:0:49:EmberEngine:Reduce:generate:37: rank=49 size=2048
generate
3564064333:0:44:EmberEngine:Reduce:generate:37: rank=44 size=2048
generate
3564146133:0:51:EmberEngine:Reduce:generate:37: rank=51 size=2048
generate
3564233133:0:46:EmberEngine:Reduce:generate:37: rank=46 size=2048
generate
3564266733:0:50:EmberEngine:Reduce:generate:37: rank=50 size=2048
generate
3564281333:0:55:EmberEngine:Reduce:generate:37: rank=55 size=2048
generate
3564362733:0:53:EmberEngine:Reduce:generate:37: rank=53 size=2048
generate
3564372333:0:52:EmberEngine:Reduce:generate:37: rank=52 size=2048
generate
3564507533:0:56:EmberEngine:Reduce:generate:37: rank=56 size=2048
generate
3564517133:0:57:EmberEngine:Reduce:generate:37: rank=57 size=2048
generate
3564569133:0:54:EmberEngine:Reduce:generate:37: rank=54 size=2048
generate
3564584333:0:59:EmberEngine:Reduce:generate:37: rank=59 size=2048
generate
3564733133:0:58:EmberEngine:Reduce:generate:37: rank=58 size=2048
generate
3564800333:0:61:EmberEngine:Reduce:generate:37: rank=61 size=2048
generate
3564809933:0:60:EmberEngine:Reduce:generate:37: rank=60 size=2048
generate
3565016333:0:62:EmberEngine:Reduce:generate:37: rank=62 size=2048
generate
3565210333:0:63:EmberEngine:Reduce:generate:37: rank=63 size=2048
generate
3565340533:0:64:EmberEngine:Reduce:generate:37: rank=64 size=2048
generate
3565434933:0:95:EmberEngine:Reduce:generate:37: rank=95 size=2048
generate
3565445333:0:67:EmberEngine:Reduce:generate:37: rank=67 size=2048
generate
3565454133:0:71:EmberEngine:Reduce:generate:37: rank=71 size=2048
generate

3565463733:0:79:EmberEngine:Reduce:generate:37: rank=79 size=2048
generate
3565475133:0:96:EmberEngine:Reduce:generate:37: rank=96 size=2048
generate
3565489133:0:68:EmberEngine:Reduce:generate:37: rank=68 size=2048
generate
3565507533:0:72:EmberEngine:Reduce:generate:37: rank=72 size=2048
generate
3565507533:0:80:EmberEngine:Reduce:generate:37: rank=80 size=2048
generate
3565522133:0:65:EmberEngine:Reduce:generate:37: rank=65 size=2048
generate
3565561133:0:111:EmberEngine:Reduce:generate:37: rank=111 size=2048
generate
3565565933:0:66:EmberEngine:Reduce:generate:37: rank=66 size=2048
generate
3565575733:0:99:EmberEngine:Reduce:generate:37: rank=99 size=2048
generate
3565594133:0:103:EmberEngine:Reduce:generate:37: rank=103 size=2048
generate
3565604333:0:112:EmberEngine:Reduce:generate:37: rank=112 size=2048
generate
3565619533:0:100:EmberEngine:Reduce:generate:37: rank=100 size=2048
generate
3565622133:0:75:EmberEngine:Reduce:generate:37: rank=75 size=2048
generate
3565631733:0:83:EmberEngine:Reduce:generate:37: rank=83 size=2048
generate
3565637933:0:104:EmberEngine:Reduce:generate:37: rank=104 size=2048
generate
3565641333:0:87:EmberEngine:Reduce:generate:37: rank=87 size=2048
generate
3565652533:0:97:EmberEngine:Reduce:generate:37: rank=97 size=2048
generate
3565665933:0:76:EmberEngine:Reduce:generate:37: rank=76 size=2048
generate
3565674933:0:84:EmberEngine:Reduce:generate:37: rank=84 size=2048
generate
3565679733:0:69:EmberEngine:Reduce:generate:37: rank=69 size=2048
generate
3565686733:0:119:EmberEngine:Reduce:generate:37: rank=119 size=2048
generate
3565690133:0:73:EmberEngine:Reduce:generate:37: rank=73 size=2048
generate
3565694733:0:88:EmberEngine:Reduce:generate:37: rank=88 size=2048
generate
3565696333:0:98:EmberEngine:Reduce:generate:37: rank=98 size=2048
generate
3565705933:0:115:EmberEngine:Reduce:generate:37: rank=115 size=2048
generate
3565708333:0:81:EmberEngine:Reduce:generate:37: rank=81 size=2048
generate
3565714733:0:70:EmberEngine:Reduce:generate:37: rank=70 size=2048
generate
3565724333:0:107:EmberEngine:Reduce:generate:37: rank=107 size=2048
generate
3565729933:0:120:EmberEngine:Reduce:generate:37: rank=120 size=2048
generate
3565733933:0:74:EmberEngine:Reduce:generate:37: rank=74 size=2048
generate
3565742733:0:91:EmberEngine:Reduce:generate:37: rank=91 size=2048
generate
3565749133:0:116:EmberEngine:Reduce:generate:37: rank=116 size=2048
generate
3565751533:0:82:EmberEngine:Reduce:generate:37: rank=82 size=2048
generate
3565767533:0:108:EmberEngine:Reduce:generate:37: rank=108 size=2048
generate
3565782733:0:113:EmberEngine:Reduce:generate:37: rank=113 size=2048
generate
3565785933:0:92:EmberEngine:Reduce:generate:37: rank=92 size=2048
generate
3565801133:0:101:EmberEngine:Reduce:generate:37: rank=101 size=2048
generate
3565816333:0:123:EmberEngine:Reduce:generate:37: rank=123 size=2048
generate

3565825933:0:114:EmberEngine:Reduce:generate:37: rank=114 size=2048
generate
3565829933:0:105:EmberEngine:Reduce:generate:37: rank=105 size=2048
generate
3565838733:0:77:EmberEngine:Reduce:generate:37: rank=77 size=2048
generate
3565844333:0:102:EmberEngine:Reduce:generate:37: rank=102 size=2048
generate
3565848333:0:85:EmberEngine:Reduce:generate:37: rank=85 size=2048
generate
3565857933:0:89:EmberEngine:Reduce:generate:37: rank=89 size=2048
generate
3565859533:0:124:EmberEngine:Reduce:generate:37: rank=124 size=2048
generate
3565873133:0:106:EmberEngine:Reduce:generate:37: rank=106 size=2048
generate
3565881933:0:78:EmberEngine:Reduce:generate:37: rank=78 size=2048
generate
3565891533:0:86:EmberEngine:Reduce:generate:37: rank=86 size=2048
generate
3565901133:0:90:EmberEngine:Reduce:generate:37: rank=90 size=2048
generate
3565912333:0:121:EmberEngine:Reduce:generate:37: rank=121 size=2048
generate
3565921933:0:117:EmberEngine:Reduce:generate:37: rank=117 size=2048
generate
3565930733:0:109:EmberEngine:Reduce:generate:37: rank=109 size=2048
generate
3565940333:0:93:EmberEngine:Reduce:generate:37: rank=93 size=2048
generate
3565955533:0:122:EmberEngine:Reduce:generate:37: rank=122 size=2048
generate
3565965133:0:118:EmberEngine:Reduce:generate:37: rank=118 size=2048
generate
3565973933:0:110:EmberEngine:Reduce:generate:37: rank=110 size=2048
generate
3565983533:0:94:EmberEngine:Reduce:generate:37: rank=94 size=2048
generate
3566032333:0:125:EmberEngine:Reduce:generate:37: rank=125 size=2048
generate
3566075533:0:126:EmberEngine:Reduce:generate:37: rank=126 size=2048
generate
3566139933:0:127:EmberEngine:Reduce:generate:37: rank=127 size=2048
generate
3566270133:0:129:EmberEngine:Reduce:generate:37: rank=129 size=2048
generate
3566486733:0:130:EmberEngine:Reduce:generate:37: rank=130 size=2048
generate
3566538133:0:131:EmberEngine:Reduce:generate:37: rank=131 size=2048
generate
3566547733:0:135:EmberEngine:Reduce:generate:37: rank=135 size=2048
generate
3566581933:0:133:EmberEngine:Reduce:generate:37: rank=133 size=2048
generate
3566591533:0:137:EmberEngine:Reduce:generate:37: rank=137 size=2048
generate
3566734133:0:143:EmberEngine:Reduce:generate:37: rank=143 size=2048
generate
3566754733:0:132:EmberEngine:Reduce:generate:37: rank=132 size=2048
generate
3566764333:0:136:EmberEngine:Reduce:generate:37: rank=136 size=2048
generate
3566782733:0:145:EmberEngine:Reduce:generate:37: rank=145 size=2048
generate
3566797933:0:134:EmberEngine:Reduce:generate:37: rank=134 size=2048
generate
3566807533:0:138:EmberEngine:Reduce:generate:37: rank=138 size=2048
generate
3566868533:0:139:EmberEngine:Reduce:generate:37: rank=139 size=2048
generate
3566903533:0:141:EmberEngine:Reduce:generate:37: rank=141 size=2048
generate
3566960333:0:144:EmberEngine:Reduce:generate:37: rank=144 size=2048
generate
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generate

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generate

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generate
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generate
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generate
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generate
3571820533:0:912:EmberEngine:Reduce:generate:37: rank=912 size=2048
generate
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generate
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generate
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generate
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generate
3571834733:0:1535:EmberEngine:Reduce:generate:37: rank=1535 size=2048
generate
3571841133:0:888:EmberEngine:Reduce:generate:37: rank=888 size=2048
generate

Bcast: ranks 2048, loop 2, bytes 56, latency 1.893 us
3571844333:0:2047:EmberEngine:Reduce:generate:37: rank=2047 size=2048
generate
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generate
3571852333:0:871:EmberEngine:Reduce:generate:37: rank=871 size=2048
generate
3571852333:0:876:EmberEngine:Reduce:generate:37: rank=876 size=2048
generate
3571853133:0:867:EmberEngine:Reduce:generate:37: rank=867 size=2048
generate
3571853133:0:963:EmberEngine:Reduce:generate:37: rank=963 size=2048
generate
3571864533:0:914:EmberEngine:Reduce:generate:37: rank=914 size=2048
generate
3571867533:0:731:EmberEngine:Reduce:generate:37: rank=731 size=2048
generate
3571868533:0:923:EmberEngine:Reduce:generate:37: rank=923 size=2048
generate
3571878133:0:660:EmberEngine:Reduce:generate:37: rank=660 size=2048
generate
3571881933:0:1071:EmberEngine:Reduce:generate:37: rank=1071 size=2048
generate
3571881933:0:806:EmberEngine:Reduce:generate:37: rank=806 size=2048
generate
3571882733:0:930:EmberEngine:Reduce:generate:37: rank=930 size=2048
generate
3571882733:0:939:EmberEngine:Reduce:generate:37: rank=939 size=2048
generate
3571885933:0:680:EmberEngine:Reduce:generate:37: rank=680 size=2048
generate
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generate
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generate
3571886933:0:863:EmberEngine:Reduce:generate:37: rank=863 size=2048
generate
3571888533:0:880:EmberEngine:Reduce:generate:37: rank=880 size=2048
generate
3571889933:0:809:EmberEngine:Reduce:generate:37: rank=809 size=2048
generate
3571896333:0:677:EmberEngine:Reduce:generate:37: rank=677 size=2048
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3571909133:0:808:EmberEngine:Reduce:generate:37: rank=808 size=2048
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3571914733:0:662:EmberEngine:Reduce:generate:37: rank=662 size=2048
generate
3571914733:0:872:EmberEngine:Reduce:generate:37: rank=872 size=2048
generate
3571915533:0:732:EmberEngine:Reduce:generate:37: rank=732 size=2048
generate
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generate
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generate
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generate
3574674733:0:1880:EmberEngine:Reduce:generate:37: rank=1880 size=2048
generate
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generate
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generate
3574678733:0:1876:EmberEngine:Reduce:generate:37: rank=1876 size=2048
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generate
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generate
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generate
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generate
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generate
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generate
3574785133:0:1855:EmberEngine:Reduce:generate:37: rank=1855 size=2048
generate
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generate
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generate
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3574894733:0:1873:EmberEngine:Reduce:generate:37: rank=1873 size=2048
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3574904333:0:1750:EmberEngine:Reduce:generate:37: rank=1750 size=2048
generate
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generate
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generate
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generate
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generate
3575017933:0:1741:EmberEngine:Reduce:generate:37: rank=1741 size=2048
generate
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generate
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generate
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generate
3575043533:0:1869:EmberEngine:Reduce:generate:37: rank=1869 size=2048
generate
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generate
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generate
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generate
3575077133:0:1719:EmberEngine:Reduce:generate:37: rank=1719 size=2048
generate
3575077133:0:1870:EmberEngine:Reduce:generate:37: rank=1870 size=2048
generate
3575080333:0:1865:EmberEngine:Reduce:generate:37: rank=1865 size=2048
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3575087533:0:1848:EmberEngine:Reduce:generate:37: rank=1848 size=2048
generate
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generate
3575097133:0:1849:EmberEngine:Reduce:generate:37: rank=1849 size=2048
generate
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generate
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generate
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generate
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generate
3575120333:0:1721:EmberEngine:Reduce:generate:37: rank=1721 size=2048
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generate
3575124333:0:1280:EmberEngine:Reduce:generate:37: rank=1280 size=2048
generate
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generate
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generate
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generate
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generate
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generate
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generate
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generate
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3575233933:0:1732:EmberEngine:Reduce:generate:37: rank=1732 size=2048
generate
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generate
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generate
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generate
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generate
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generate
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generate
3575330733:0:1282:EmberEngine:Reduce:generate:37: rank=1282 size=2048
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generate
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generate
3575431533:0:1854:EmberEngine:Reduce:generate:37: rank=1854 size=2048
generate
3575492333:0:1716:EmberEngine:Reduce:generate:37: rank=1716 size=2048
generate
3575527533:0:1718:EmberEngine:Reduce:generate:37: rank=1718 size=2048
generate
3575595533:0:1724:EmberEngine:Reduce:generate:37: rank=1724 size=2048
generate
3575642733:0:1726:EmberEngine:Reduce:generate:37: rank=1726 size=2048
generate
3575892333:0:1793:EmberEngine:Reduce:generate:37: rank=1793 size=2048
generate
3575901933:0:1792:EmberEngine:Reduce:generate:37: rank=1792 size=2048

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generate
3576108333:0:1794:EmberEngine:Reduce:generate:37: rank=1794 size=2048
generate
3576493133:0:1408:EmberEngine:Reduce:generate:37: rank=1408 size=2048
generate
3576554733:0:1920:EmberEngine:Reduce:generate:37: rank=1920 size=2048
generate
3576797133:0:1664:EmberEngine:Reduce:generate:37: rank=1664 size=2048
Allgather: ranks 2048, loop 3, bytes 12, latency 62.161 us
Barrier: ranks 2048, loop 99, latency 38.979 us
7667251533:0:0:EmberEngine:PingPong:generate:65: rank=0 size=2048
rank2=1 messageSize=256 iterations=35
7668094733:0:1:EmberEngine:PingPong:generate:65: rank=1 size=2048
7672983533:0:32:EmberEngine:PingPong:generate:65: rank=32 size=2048
7681623533:0:2047:EmberEngine:PingPong:generate:65: rank=2047 size=2048
7682296333:0:1024:EmberEngine:PingPong:generate:65: rank=1024 size=2048
PingPong: otherRank 1, total time 63.474 us, loop 35, bufLen 256, latency 0.907 us. bandwidth 0.282320 G
B/s
7730725133:0:0:EmberEngine:PingPong:generate:65: rank=0 size=2048
rank2=32 messageSize=256 iterations=35
PingPong: otherRank 32, total time 159.992 us, loop 35, bufLen 256, latency 2.286 us. bandwidth 0.112006
GB/s
7890717133:0:0:EmberEngine:PingPong:generate:65: rank=0 size=2048
rank2=1024 messageSize=256 iterations=35
PingPong: otherRank 1024, total time 159.992 us, loop 35, bufLen 256, latency 2.286 us. bandwidth 0.1120
06 GB/s
8050709133:0:0:EmberEngine:PingPong:generate:65: rank=0 size=2048
rank2=2047 messageSize=256 iterations=35
PingPong: otherRank 2047, total time 438.200 us, loop 35, bufLen 256, latency 6.260 us. bandwidth 0.0408
95 GB/s
Simulation is complete, simulated time: 8.51708 ms
All PPN=2 mesh runs finished. Results in: /Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/Mini
FE/Mesh/PPN_2/results
stephenrajva@Stephens-MacBook-Air-4 PPN_2 % cd "/Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulato
r/MiniFE/Mesh/PPN_2"
ls -l results/*.csv
-rw-r--r--  1 stephenrajva  staff   989312 28 Jun 06:18 results/mesh_16x16_ppn2_nodes512_stats.csv
-rw-r--r--@ 1 stephenrajva  staff   12095 28 Jun 06:18 results/mesh_2x2_ppn2_nodes8_stats.csv
-rw-r--r--  1 stephenrajva  staff  4068966 28 Jun 06:19 results/mesh_32x32_ppn2_nodes2048_stats.csv
-rw-r--r--  1 stephenrajva  staff   55243 28 Jun 06:18 results/mesh_4x4_ppn2_nodes32_stats.csv
-rw-r--r--  1 stephenrajva  staff  236080 28 Jun 06:18 results/mesh_8x8_ppn2_nodes128_stats.csv
stephenrajva@Stephens-MacBook-Air-4 PPN_2 % cd "/Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulato
r/MiniFE/Mesh/PPN_2"
echo "==== PPN=2 CSVs ====="
ls -l results/*.csv
echo ""
echo "==== finding analyze_universal.py ====="
find "/Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/MiniFE/Mesh" -name "analyze_universal.py
" 2>/dev/null
==== PPN=2 CSVs =====
-rw-r--r--  1 stephenrajva  staff   989312 28 Jun 06:18 results/mesh_16x16_ppn2_nodes512_stats.csv
-rw-r--r--@ 1 stephenrajva  staff   12095 28 Jun 06:18 results/mesh_2x2_ppn2_nodes8_stats.csv
-rw-r--r--  1 stephenrajva  staff  4068966 28 Jun 06:19 results/mesh_32x32_ppn2_nodes2048_stats.csv
-rw-r--r--  1 stephenrajva  staff   55243 28 Jun 06:18 results/mesh_4x4_ppn2_nodes32_stats.csv
-rw-r--r--  1 stephenrajva  staff  236080 28 Jun 06:18 results/mesh_8x8_ppn2_nodes128_stats.csv

==== finding analyze_universal.py =====
/Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/MiniFE/Mesh/ppn1/analyze_universal.py
stephenrajva@Stephens-MacBook-Air-4 PPN_2 % cd "/Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulato
r/MiniFE/Mesh/PPN_2"
cp ../ppn1/analyze_universal.py .
ls -l analyze_universal.py
-rw-r--r--@ 1 stephenrajva  staff   6965 28 Jun 06:24 analyze_universal.py
stephenrajva@Stephens-MacBook-Air-4 PPN_2 % cd "/Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulato
r/MiniFE/Mesh/PPN_2"
python3 - << 'EOF'
import re
f = "analyze_universal.py"
src = open(f).read()

# New config for the PPN=2 mesh study
src = re.sub(r'^STUDY\s*=\s*$', 'STUDY = "MiniFE_Mesh_PPN2"', src, count=1, flags=re.M)
src = re.sub(r'^TOPOLOGY\s*=\s*$', 'TOPOLOGY = "mesh"', src, count=1, flags=re.M)

runs = '''RUNS = [
    ("2x2",    (2, 2),    "results/mesh_2x2_ppn2_nodes8_stats.csv"),

```